COE Applied AI Workshop

Wednesday July 15

Time	Activity	Lead
1:00 pm - 2:00 pm	From Engineering or Disciplinary Problem to Machine Learning Problem: defining the prediction objective, inputs, outputs, unit of observation, target variables, baselines, and success criteria	Fred Livingston
2:00 pm - 2:30 pm	Classic Machine Learning Overview: regression, classification, clustering, dimensionality reduction, forecasting, and anomaly detection	Fred Livingston
2:30 pm - 3:00 pm	Presentation: From Prompts to Agents: Scalable, Efficient, and Grounded Agentic AI for Engineering Workflows	Dongkuan (DK) Xu
	Dongkuan (DK) Xu is an Assistant Professor of Computer Science at NC State and leads the Generative Intelligent Computing (GIC) Lab. His research develops scalable, efficient, and grounded agentic AI systems that reason, use tools, and operate in real-world scientific and engineering workflows. His broader research spans deep learning, large language models, computer vision, and applications in scientific discovery, environmental intelligence, education, robotics, networking, and healthcare. He has received honors including the NC State Dean's Applied AI Research Accelerator Award, AAAI Best Demo Award, AAAI New Faculty Highlights recognition, NVIDIA Academic Grant Program Award, and Microsoft Accelerating Foundation Models Research Award.	
3:00 pm - 3:15 pm	Coffee Break	
3:15 pm - 3:45 pm	Regression Fundamentals: linear regression, polynomial regression, regularization, residuals, and regression metrics	Fred Livingston
3:45 pm - 4:15 pm	Classification Fundamentals: logistic regression, decision trees, confusion matrices, precision, recall, and F1-score	Fred Livingston
4:15 pm - 5:00 pm	Hands-On Activity: Building a Baseline ML Model: prepare a small dataset, establish a baseline, train regression or classification models, and interpret initial results	Fred Livingston

Thursday July 16

Time	Activity	Lead
1:00 pm - 2:00 pm		

2:00 pm - 2:30 pm		
2:30 pm - 3:00 pm	Presentation: LLM and AI Workflows on NCSU's Hazel HPC	Andrew Petersen
	Dr. Andrew Petersen is a Computer Scientist at NC State University's High Performance Computing (HPC) center, and the Principal Investigator for a 2023–2025 NSF award focused on expanding campus GPU capacity for AI and data-intensive research. He leads university efforts to deploy local, scalable LLM and RAG services on HPC infrastructure. He also lectures for the Data Science & AI Academy, where he works to democratize large-scale analytics by deploying Apache Spark for use across all academic disciplines.	
3:00 pm - 3:15 pm	Coffee Break	
3:15 pm - 3:45 pm		
3:45 pm - 4:15 pm		
4:15 pm - 5:00 pm		
Friday July 17		
Time	**Activity**	**Lead**
1:00 pm - 2:00 pm		
2:00 pm - 2:30 pm		
2:30 pm - 3:00 pm	Presentation: NeuralSmith: An Automated AI/ML Engineering Co-Pilot for Domain Scientists and Engineers	Vijay K. Shah
	Vijay K. Shah, Ph.D. is an Assistant Professor in the Department of Electrical and Computer Engineering at NC State University, where he directs the NextG Wireless Lab. His research spans AI-native wireless networks, Open RAN, intelligent spectrum management, and the application of LLMs and multi-agent AI to telecommunications infrastructure, with support from NSF, NIST, NTIA, and NASA. He is a recipient of the NSF CAREER Award (2025) and the Goodnight Early Career Innovators Award (2025-26). He serves as Outreach Director of AERPAW, the NSF PAWR wireless research platform, and is Co-Founder of WiSights Lab and All Things Intelligence Inc., the company behind NeuralSmith. He co-authored Fundamentals of O-RAN (IEEE Press/Wiley) and is a Senior Member of the IEEE.	
3:00 pm - 3:15 pm	Coffee Break	
3:15 pm - 3:45 pm		
3:45 pm - 4:15 pm		
4:15 pm - 5:00 pm		